4D Cluster Density Analysis

Information Density in K-map Minimization

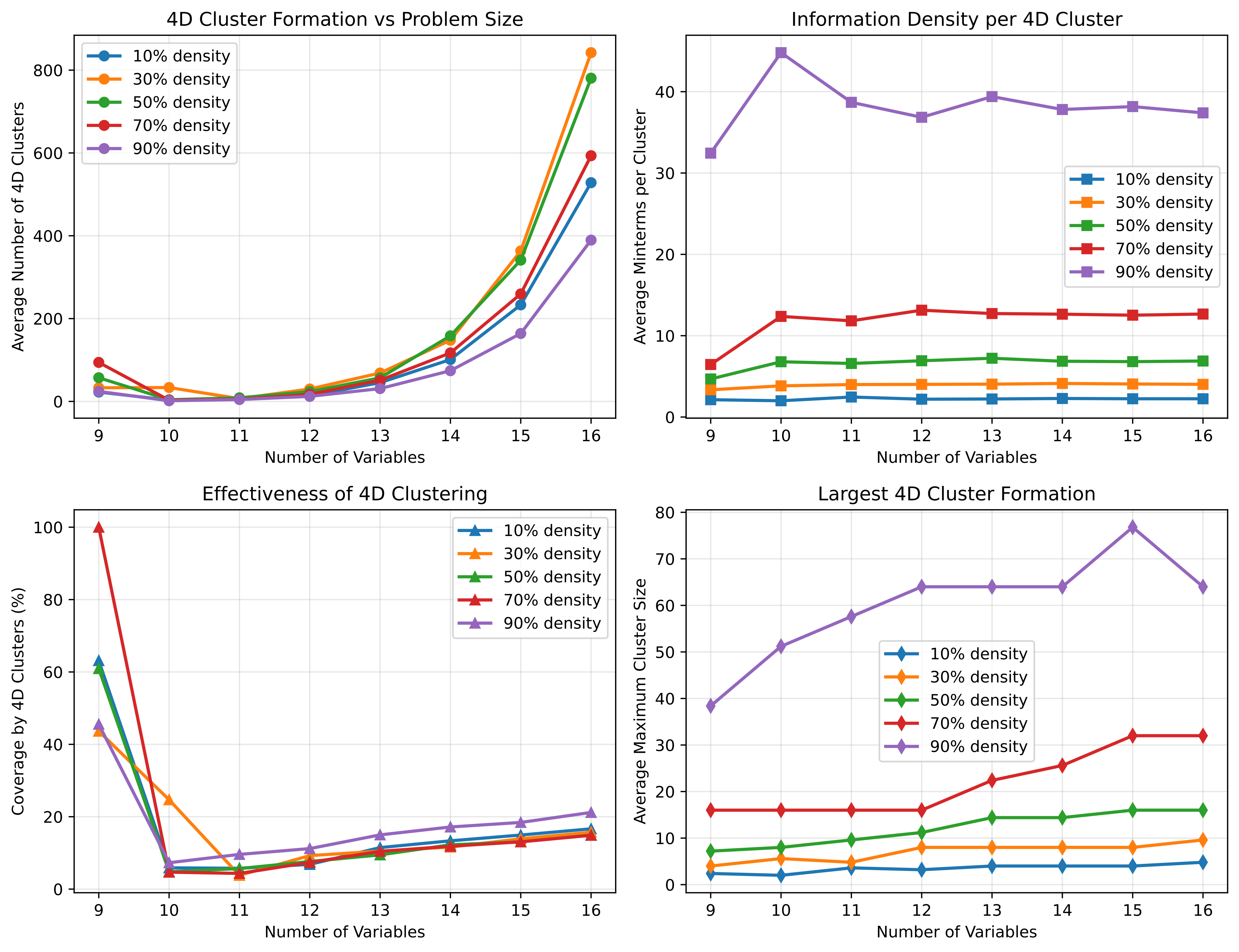
Analysis Date: 2026-01-19

Random Seed: 42

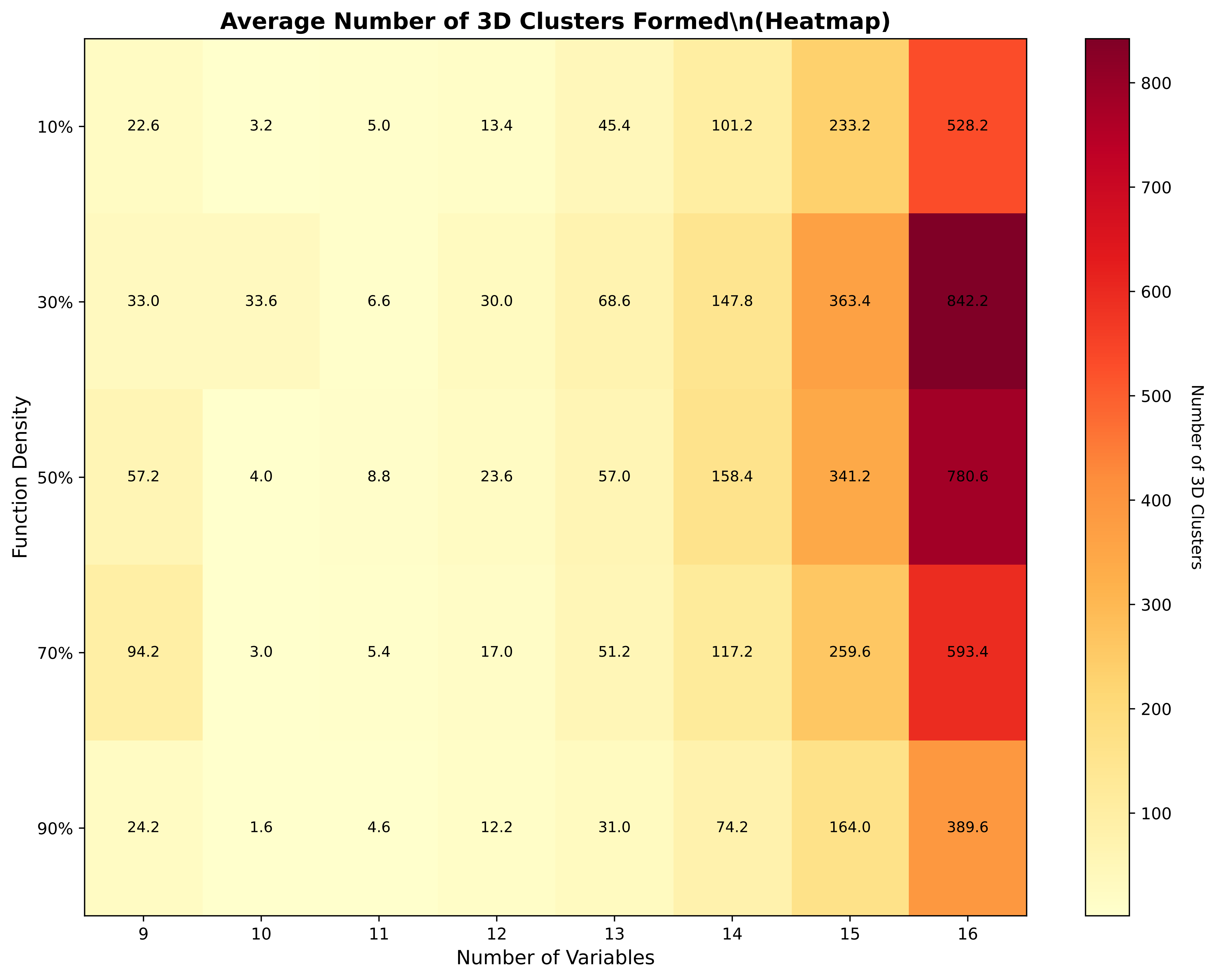
Total Test Cases: 200

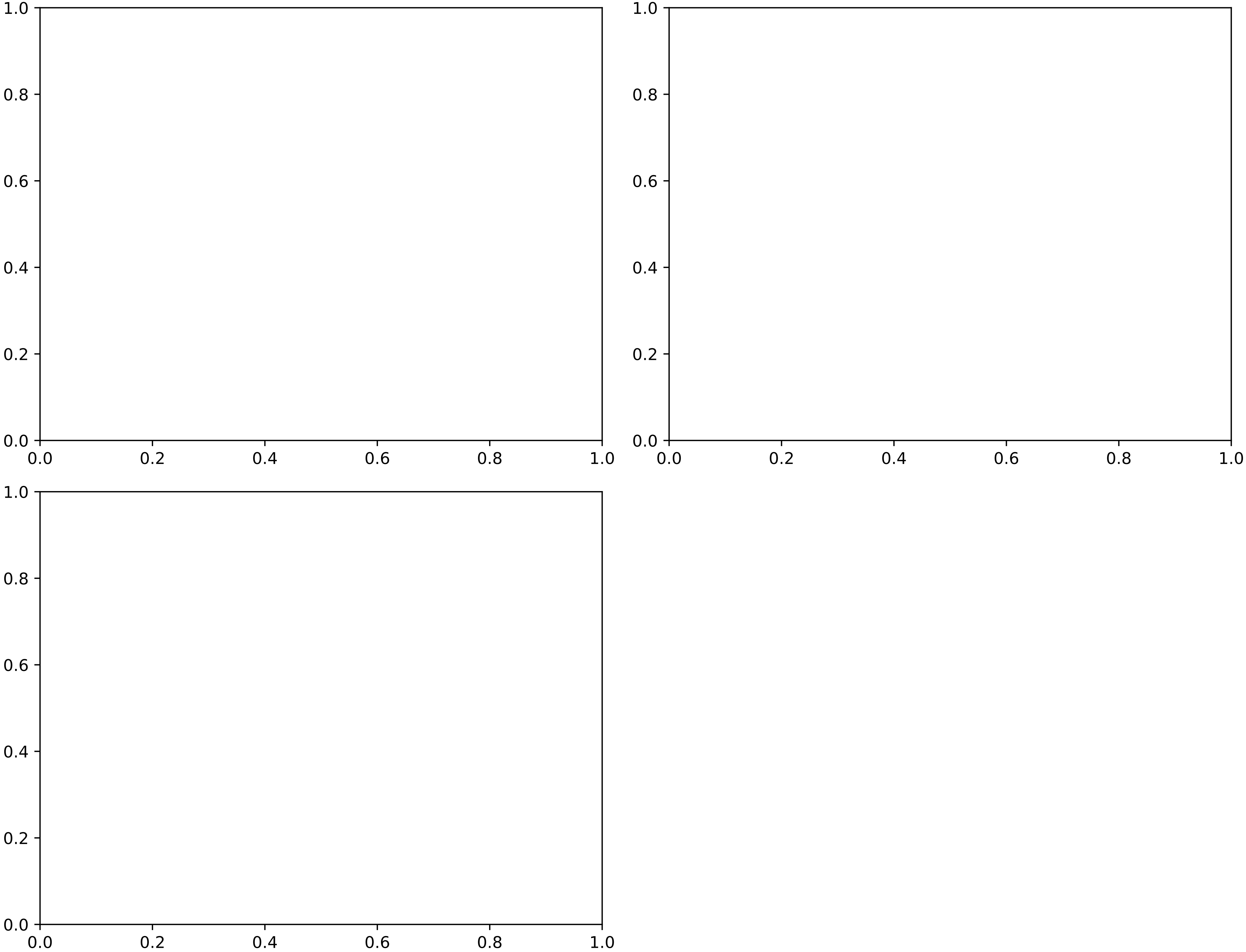
Variable Range: 9-16

Geometric Clustering Behavior Analysis



4D Cluster Formation vs Problem Size
Average Number of 4D Clusters
Number of Variables
Information Density per 4D Cluster
Average Minterms per Cluster
Number of Variables
Effectiveness of 4D Clustering
Coverage by 4D Clusters (%)
Number of Variables
Largest 4D Cluster Formation
Average Maximum Cluster Size
Number of Variables
10% density
30% density
50% density
70% density
90% density


Average Number of 3D Clusters Formed\n(Heatmap)
Function Density
10%
30%
50%
70%
90%
Number of Variables
9
10
11
12
13
14
15
16
22.6 3.2 5.0 13.4 45.4 101.2 233.2 528.2
33.0 33.6 6.6 30.0 68.6 147.8 363.4 842.2
57.2 4.0 8.8 23.6 57.0 158.4 341.2 780.6
94.2 3.0 5.4 17.0 51.2 117.2 259.6 593.4
24.2 1.6 4.6 12.2 31.0 74.2 164.0 389.6
Number of 3D Clusters
100
200
300
400
500
600
700
800


0.0
0.2
0.4
0.6
0.8
1.0

Analysis Summary

Key Findings

- Total tests conducted: 200
- Average 4D clusters per test: 143.76
- Average coverage by 4D clusters: 17.8%
- Variable range: 9 to 16
- Density levels tested: 5

Observations

- *4D cluster formation increases with variable count*
- *Higher density functions tend to form more 4D clusters*
- *Average cluster size varies with function structure*
- *Coverage by 4D clusters is significant in most cases*

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